1 ILP Formulation

Model

$$\min \quad C_{\max} + \lambda_{\text{core}} \sum_{c \in C} o_c^{\text{core}} + \lambda_{\text{cluster}} \sum_{k \in K} o_k^{\text{cluster}} + \sum_{(i,j) \in D} \sum_{c \in E_i} \sum_{d \in E_j} q_{cd} \, z_{ijcd} \tag{1}$$

Subject to:

Assignment

$$\sum_{c \in E_i} x_{ic} = 1 \qquad \forall i \in T \tag{2}$$

$$x_{ic} = 0 \qquad \forall i \in T,\ c \notin E_i \tag{3}$$

Timing

$$s_i \geq r_i \qquad \forall i \in T \tag{4}$$

$$f_i = s_i + \sum_{c \in C} p_i \alpha_c x_{ic} \qquad \forall i \in T \tag{5}$$

$$C_{\max} \geq f_i \qquad \forall i \in T \tag{6}$$

Precedence

$$s_j \geq f_i \qquad \forall (i,j) \in D \tag{7}$$

Core memory

$$\sum_{i \in T} m_i x_{ic} \leq B_c^{\text{core}} + o_c^{\text{core}} \qquad \forall c \in C \tag{8}$$

Cluster memory

$$\sum_{i \in T} \sum_{c \in C_k} m_i x_{ic} \leq B_k^{\text{cluster}} + o_k^{\text{cluster}} \qquad \forall k \in K \tag{9}$$

Non-overlap (same core)

$$s_j \geq f_i - M(3 - y_{ij} - x_{ic} - x_{jc}) \qquad \forall i < j,\ c \in C \tag{10}$$

$$s_i \geq f_j - M(2 + y_{ij} - x_{ic} - x_{jc}) \qquad \forall i < j,\ c \in C \tag{11}$$

Communication linearization

$$z_{ijcd} \leq x_{ic} \qquad \forall (i,j) \in D,\ c \in E_i,\ d \in E_j \tag{12}$$

$$z_{ijcd} \leq x_{jd} \qquad \forall (i,j) \in D,\ c \in E_i,\ d \in E_j \tag{13}$$

$$z_{ijcd} \geq x_{ic} + x_{jd} - 1 \qquad \forall (i,j) \in D,\ c \in E_i,\ d \in E_j \tag{14}$$

Domains

$$x_{ic},\ y_{ij},\ z_{ijcd} \in \{0,1\} \tag{15}$$

$$s_i,\ f_i,\ C_{\max} \geq 0 \tag{16}$$

$$o_c^{\text{core}},\ o_k^{\text{cluster}} \geq 0 \tag{17}$$